Vanderbilt
Social
Good
Hackathon

ShowSignal

Reducing patient no-shows by bridging

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About Me

CS + Math ‘27

Passionate about full-stack, AI & health-tech -> previously at Stanford Radiology & Amazon

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No-Show Impact

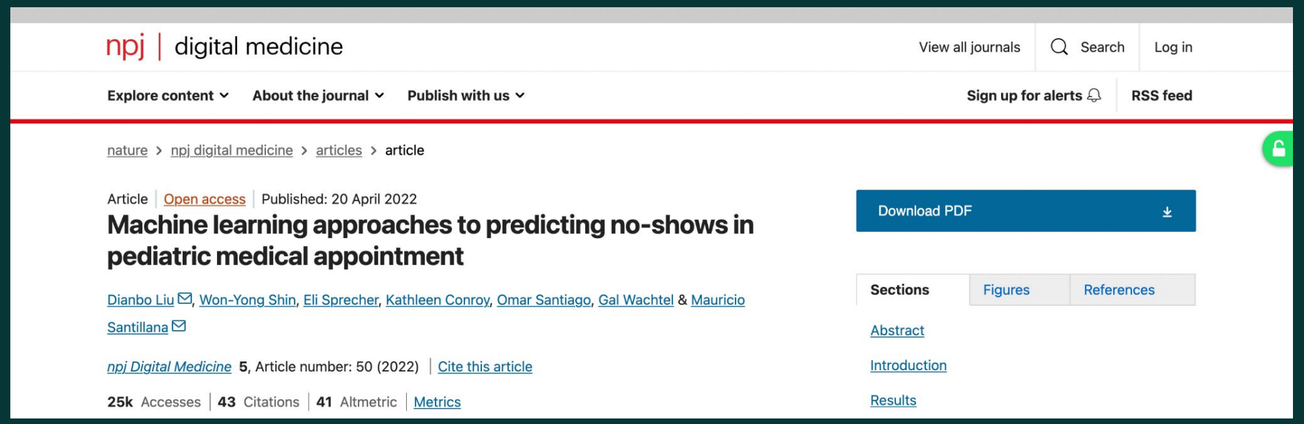
15-30% of appointments made in the US are **no-shows**

Wasted resources & time cause:

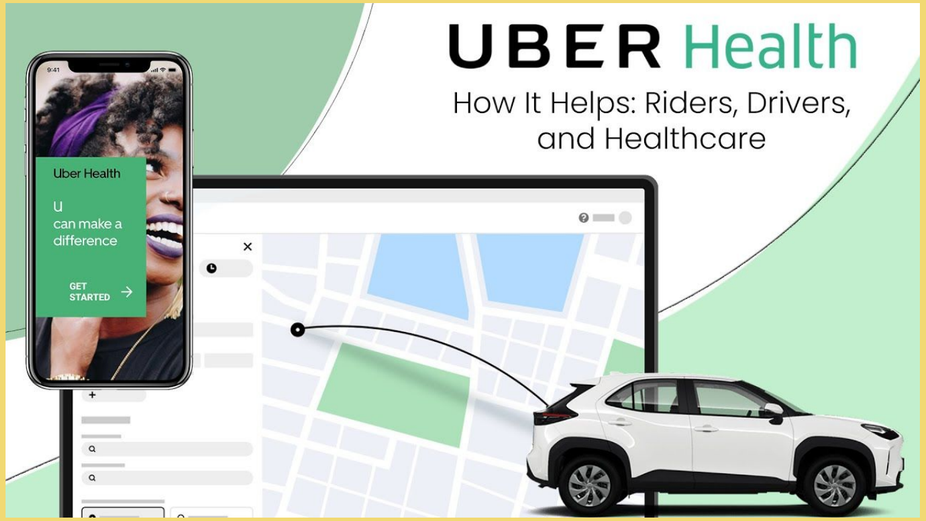
- **Tens of billions** of dollars of lost
- Longer wait times for everyone else
- Patients miss out on care

This number **exceeds 30-40%** for patients from underserved populations, due to barriers like

- Transport
- Language
- Financial
- Taking time off work, arranging childcare, etc



Motivations



Observed market mismatch

Waste physician time and cause administrative waste
VUMC & C4E faculty describe it as a huge problem
Services exist … but enough people don't know about them
→ **The opportunity:** use formal hospital systems to promote existing services and help more patients make their appointments

Personal experience + context

VUMC is a well-funded, leading hospital
However, Southern residents are more likely to:

- be below poverty
- have transit issues (poor infrastructure)
- issues navigating legal/insurance

No-Show Risk Prediction

Dataset from a clinic in Brazil studying no shows

- Disability (intellectual or motor), whether patient needs companion, distance from clinic, weather/time of appointment

ML Model

Trained **SKLearn random forest model** to predict patients as high, medium, low risk for no-show

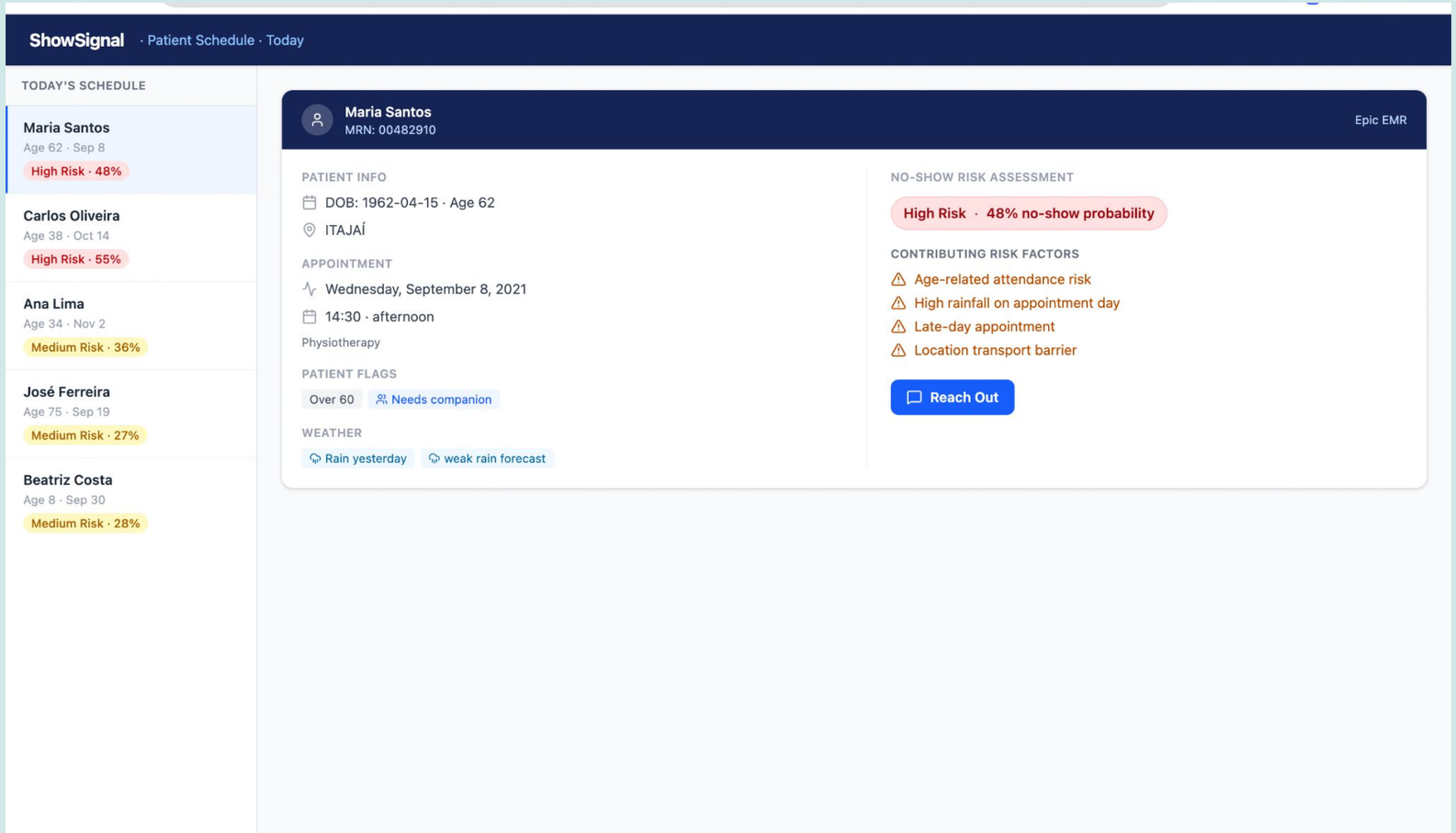
Risk Factors

Identified variables that cause high risk such as old age, physical disability, etc

FastAPI Endpoints

Serves patient risk and high-risk variables to be rendered on clinician dashboard in frontend

Frontend



Built with **React, TailwindCSS** to mock real Epic systems

Reach Out buttons only exist for high & medium-risk patients (5 were selected)

Minimize clinician burden (simple button)

Conversational Agent

Harnessing AI in medicine in a way that is both **safe/lower-risk** than for clinical advising and emphasizes **personalized** connection

Context-building

Doctor clicks “Reach Out” - agent builds context

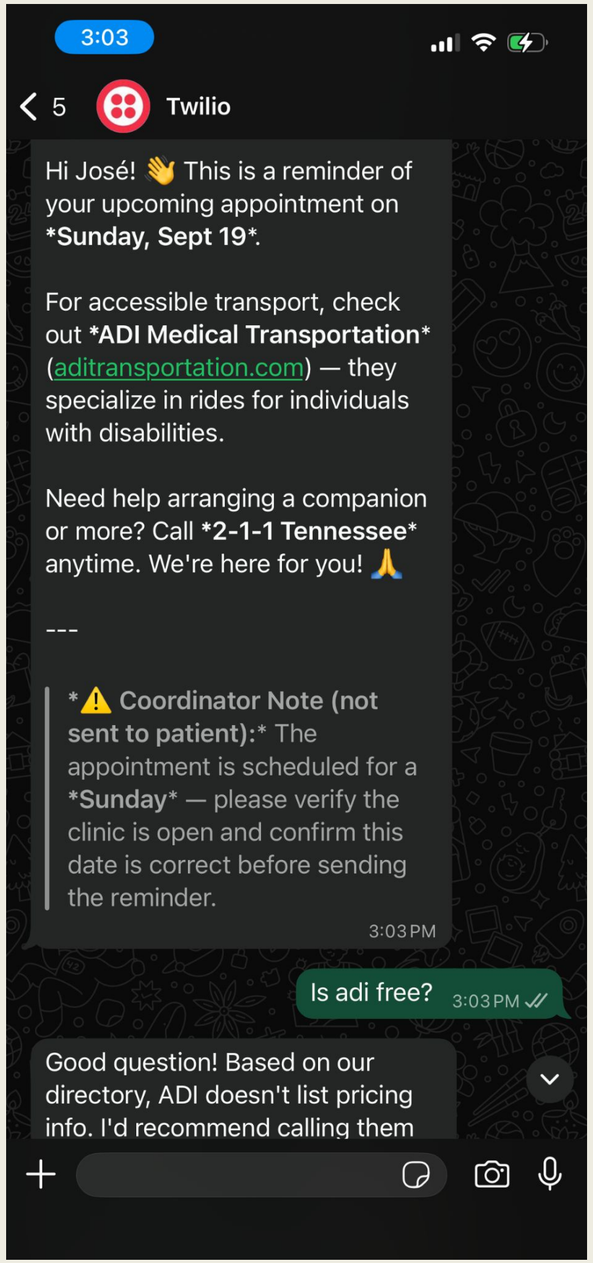
- looks through patient data record
- matches their flags with appropriate resources in knowledge base

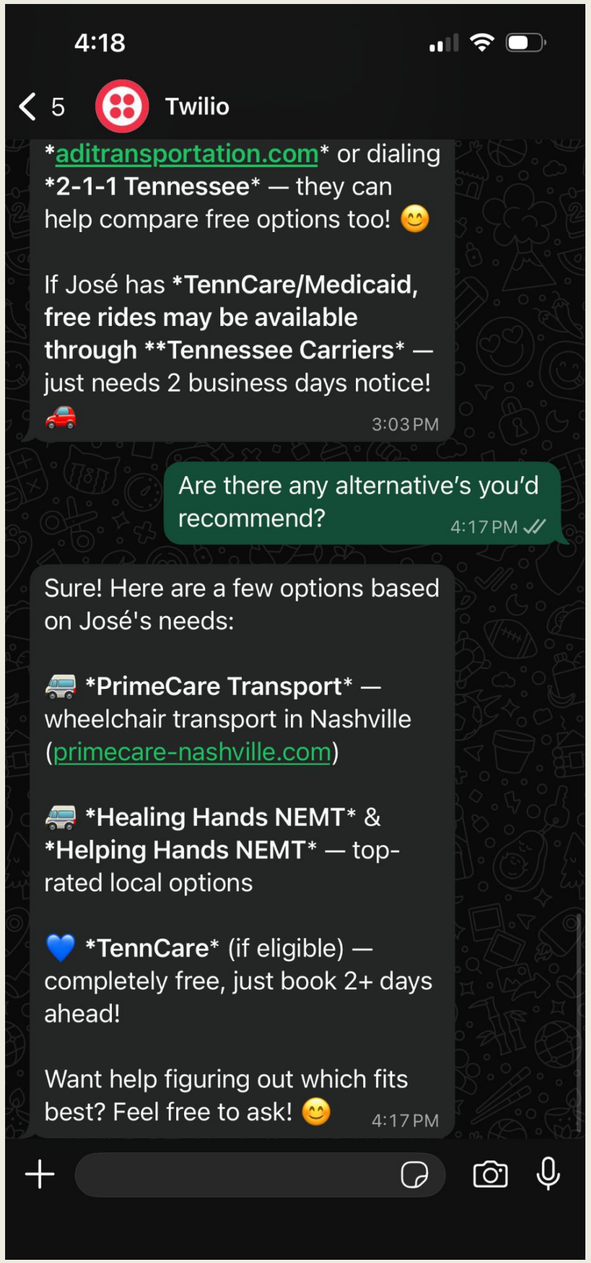
Initial Outreach

Friendly, personalized note recommending resources
Sent using **Twilio API** using **WhatsApp**
All convos stored by memory, keyed by phone number

Dynamic Conversation

Using **ngrok webhooks**, incoming messages from the patient are passed to **Claude agent**
Can search the web & refer to knowledge base to answer questions and **provide assistance**





Example

Focus on providing actionable and critical information

Future Direction - Adapting to Epic

Designed with future real clinic interfaces in mind

Epic CDS Hooks

Backend **REST API** for decision support is compatible with how Epic CDS Hooks function

FHIR Health Data APIs

Epic has **FHIR APIs** for calling patient data fields that were used for risk prediction

UI Add-On

Small buttons/flags can be integrated as a **SmartApp** in Epic frontend

Communication Channels

EPIC has frameworks for sending text, email, MyChart messages to patients -> integrate

Thank you!

Free Q&A time if available :)